

NAIRR Pilot

National AI Research Resource — lightweight path to DGX Cloud, Azure, and DOE GPUs

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Program Overview

The National Artificial Intelligence Research Resource (NAIRR) Pilot launched in January 2024 as a coordinated two-year demonstration project led by the U.S. National Science Foundation (NSF) in partnership with the Department of Energy (DOE), the National Institutes of Health (NIH), DARPA, NASA, NIST, and a broad consortium of industry and academic providers. The Pilot is the precursor to a full-scale, statutorily authorized NAIRR that the administration and Congress are now moving to stand up as a permanent program. NSF released the NAIRR Operations Center (NAIRR-OC) solicitation in late 2025 with a full proposal deadline of February 4, 2026, signaling the transition from pilot to permanence. The Pilot itself is funded and provisioned to operate through approximately mid-2026, with allocations remaining open until provider resources are exhausted.

The provider list is unusually broad for a federal compute program. On the commercial side, NVIDIA contributes roughly \$30M of value, including \$24M in DGX Cloud credits (H100 and H200 systems). Microsoft contributes \$20M in Azure credits across ND H100 v5 and ND H200 v5 families, plus named-model access. Amazon Web Services, Google Cloud, OpenAI, Anthropic, Hugging Face, Meta, IBM, Cerebras, Groq, and several other vendors provide either credits, model access, or hosted inference. On the public side, Argonne National Laboratory (Polaris, Aurora, AI Testbed), Oak Ridge National Laboratory, Lawrence Berkeley National Laboratory's NERSC (Perlmutter), TACC (Vista, Lonestar6), PSC (Bridges-2, Neocortex), NCSA (Delta, DeltaAI), SDSC (Expanse), Purdue (Anvil), and Indiana University (Jetstream-2) all contribute capacity. The Pilot's scientific scope is deliberately AI/ML-only, in contrast to ACCESS, which is general HPC.

Four Call Types

The Pilot is organized around four primary calls plus a Deep Partnerships track. The structure rewards small, fast asks for routine work and reserves heavier review for the largest grants.

Call	Purpose	Term	Cadence
Classroom	Course-integrated AI/ML instruction with shared student accounts and curated datasets.	1 semester	Rolling, 2 weeks
Research	Open-ended AI/ML research at the project level; the workhorse track for grad students and PIs.	12 months	Rolling, 3–4 weeks
Education	Workshops, tutorials, summer schools, REUs, and broader training events.	Event-bound	Rolling, 2–3 weeks

Call	Purpose	Term	Cadence
Software	Tooling, benchmarks, model hubs, and infrastructure that the broader NAIRR community will reuse.	12 months	Rolling, 3–4 weeks

A separate Start-Up call provides 3-month, capped allocations (approximately 2,000 GPU-hours on TACC Vista, Purdue Anvil AI, SDSC Expanse AI, or PSC Bridges-2 GPU; 500 node-hours on the ANL AI Testbed; or 50 CS-2 hours on PSC Neocortex) with a turnaround inside three weeks. Start-Up is the recommended on-ramp for new users, and a successful Start-Up project makes a follow-on Research request straightforward.

Hardware Specs

The table below lists the public, AI-relevant accelerators reachable via a NAIRR allocation. Aurora is included for completeness even though its Intel Max 1550 (Ponte Vecchio) GPUs are not CUDA-compatible and require oneAPI or SYCL ports. The “TFLOPS” column is dense FP16/BF16 tensor throughput without sparsity, so figures are directly comparable to the 989 TFLOPS H200 baseline used elsewhere in this report.

Resource	GPU	VRAM (GB)	FP16 TFLOPS	BW (TB/s)	Interconnect
NVIDIA DGX Cloud	H100 SXM	80	989	3.35	NVLink 4 + 3.2 Tbps IB
NVIDIA DGX Cloud	H200 SXM	141	989	4.80	NVLink 4 + 3.2 Tbps IB
Azure ND H100 v5	H100 SXM	80	989	3.35	NVLink 4 + 1.6 Tbps IB
Azure ND H200 v5	H200 SXM	141	989	4.80	NVLink 4 + 3.2 Tbps IB
ANL Polaris	A100 40GB	40	312	1.55	HPE Slingshot 11
ANL Aurora ¹	Intel Max 1550	128	419	3.30	HPE Slingshot 11
NERSC Perlmutter	A100 40GB	40	312	1.55	HPE Slingshot 10
NCSA Delta	A100 40GB	40	312	1.55	HPE Slingshot 10
NCSA DeltaAI	GH200	96	989	4.00	HPE Slingshot 11
Purdue Anvil AI	A100 40GB	40	312	1.55	HDR-200 InfiniBand

¹Intel Max 1550 GPUs require oneAPI/SYCL; CUDA code does not run unmodified.

Resource	GPU	VRAM (GB)	FP16 TFLOPS	BW (TB/s)	Interconnect
TACC Vista	GH200	96	989	4.00	NDR-400 In- finiBand

Allocation Sizes

NAIRR allocations are denominated inconsistently across providers. Cloud partners issue dollar credits, DOE labs issue node-hours, and NSF systems issue service units (SUs) or GPU-hours that map to ACCESS conventions. The table below normalizes typical Research-call awards into *H200-hour equivalents* using dense FP16 throughput. Conversions are $\text{hours} \times \frac{\text{TFLOPS}_{\text{system}}}{989}$, summed over the per-project award size, with cloud credits converted at on-demand list pricing for the relevant SKU.

Resource	Typical award	Native units	H200-hr equivalent
NVIDIA DGX Cloud	\$25k credit (H100/H200)	640 GPU-hr	640
Azure ND H200 v5	\$5k–\$80k tiered	125–2,000 GPU-hr	125–2,000
Azure ND H100 v5	\$5k–\$80k tiered	150–2,400 GPU-hr	150–2,400
ANL Polaris	25,000 node-hr (4×A100)	100,000 GPU-hr	31,500
NERSC Perlmutter	50,000 GPU-hr	50,000 GPU-hr	15,800
NCSA Delta (A100)	40,000 GPU-hr	40,000 GPU-hr	12,600
NCSA DeltaAI (GH200)	10,000 GPU-hr	10,000 GPU-hr	10,000
Purdue Anvil AI	10,000 GPU-hr	10,000 GPU-hr	3,160
TACC Vista (GH200)	10,000 GPU-hr	10,000 GPU-hr	10,000
ANL Aurora	5,000 node- hr (6×Max1550)	30,000 GPU-hr	12,700

Worked Example

A 50,000 GPU-hour Perlmutter A100 award contributes $50000 \times \frac{312}{989} \approx 15780$ H200-hours of dense FP16 compute. The same project supplemented by a \$25k DGX Cloud H200 credit (about 640 H200-hours at the published reservation rate of roughly \$3.90/GPU-hr) and a 10,000 GPU-hour DeltaAI GH200 award yields roughly $15780 + 640 + 10000 = 26420$ H200-hours total. That is enough to train a 13B-parameter dense transformer from scratch on 300B tokens at industry-standard utilization, or to run several thousand fine-tuning sweeps on 7B-class models.

Approval Timeline

NAIRR review cadence is one of its main competitive advantages. Start-Up projects are reviewed continuously and decided within three weeks. Research allocations are reviewed monthly, with most awards posted two to four weeks after submission. Classroom requests are typically turned around in under two weeks, and Education event allocations are reviewed against the event date. The exception is *DGX Cloud*,

which is heavily oversubscribed; the review committee meets monthly but the queue can stretch several months when demand spikes, and not every meritorious request is funded on the first cycle. Azure ND H200 v5 has been the most consistently available high-end tier through 2025–2026.

The application itself is short. The standard Research request is a 2–3 page project description with no required letters of support, no required faculty sponsor for postdocs, and a one-page methods/compute-justification section. Graduate students may apply with a PI sponsor signature; the PI need not be the requester. There is no cost share and no indirect cost recovery.

Eligibility

Eligibility is broader than ACCESS but narrower than purely commercial credit programs. Requesters must be affiliated with a U.S.-based academic, nonprofit, or federal research institution, or with a U.S. small business. Graduate students and postdocs are eligible directly with PI sponsorship. Non-U.S. collaborators may be added to a project but cannot be the lead PI, and certain partner resources (notably DGX Cloud and some DOE testbeds) require additional citizenship or institutional screening before account activation. Projects involving export-controlled data, classified work, or human-subjects data carrying HIPAA obligations face additional review and are routed only to compatible providers.

Things to Know

Several practical points materially affect how to use NAIRR. *DGX Cloud* and *Azure* behave very differently in practice. DGX Cloud is requested as a dollar credit, drawn down on a managed reservation, and the review committee actively allocates against a fixed pool, so framing the request around throughput-per-experiment rather than wall-clock helps. Azure credits are spent against on-demand or reserved SKUs, and underused credits expire at the end of the 12-month term with no rollover, so plan a burn schedule up front.

Storage is generally *not* part of the allocation. DOE labs provide scratch and project filesystems sized to the compute award, but cloud providers expect researchers to provision and pay for their own object storage out of the same credit pool, which can quietly consume 10–20% of an Azure award on multi-terabyte datasets. NAIRR also imposes an open-science requirement. Resulting publications, code, and (where licensing permits) trained model artifacts must be deposited in a public repository and acknowledged with the standard NAIRR citation.

NAIRR allocations stack with ACCESS allocations, and there is no penalty for holding both. The recommended pattern is to use ACCESS Explore or Discover for general HPC and CPU-heavy preprocessing, NAIRR Research for H100/H200/GH200 training, and NAIRR Start-Up to bridge the gap during ACCESS review.

Strategic Fit

For a graduate AI researcher who needs Hopper-class GPUs in weeks rather than months, NAIRR is the fastest legitimate path. The application is roughly one-third the length of an ACCESS Accelerate

request, requires no letters, and is reviewed monthly against an AI-only docket where ML projects are not competing against general HPC. Recommend NAIRR Start-Up as the first-line application for any new project; promote to NAIRR Research once preliminary results justify a 12-month award; and request DGX Cloud or Azure ND H200 v5 only when a documented capability gap (FP8 training, NVLink-4 all-reduce bandwidth, 141 GB VRAM for activation memory) makes the on-prem A100 systems inadequate.

Summary

Resource	Approval	Typical size	H200-hr eq.	Ease
NAIRR Start-Up	< 3 wk	2,000 GPU-hr	640–2,000	Very high
Research / Anvil A100	3–4 wk	10,000 GPU-hr	3,160	High
Research / Perlmutter	3–4 wk	50,000 GPU-hr	15,800	High
Research / Polaris	3–4 wk	25,000 node-hr	31,500	High
Research / DeltaAI GH200	3–4 wk	10,000 GPU-hr	10,000	Medium
Research / TACC Vista	3–4 wk	10,000 GPU-hr	10,000	Medium
Research / Azure H200	3–4 wk	\$5k–\$80k	125–2,000	Medium
Research / DGX Cloud	4–12 wk	\$25k credit	640	Low (queue)
Aurora (Intel Max)	3–4 wk	5,000 node-hr	12,700	Low (port)

H200-hour equivalence is computed at dense FP16/BF16 tensor throughput (989 TFLOPS, 141 GB HBM3e, 4.8 TB/s) per H200 SXM GPU. Conversions below use the per-hour FP16 TFLOP ratio; memory-bound workloads scale differently and the bandwidth column is provided so readers can re-derive their own equivalence.

Sources

- NAIRR Pilot home and allocations catalog: nairrpilot.org · [/allocations](#) · [/opportunities/allocations](#) · [/opportunities/startup-project](#)
- NAIRR Pilot proposal instructions and FAQ: [proposal instructions](#) · [FAQ](#)
- NAIRR Deep Partnerships and Azure Deep Partnership: [NAIRR Deep Partnerships](#) · [CloudBank Azure Deep Partnership](#)
- NSF NAIRR program page and 2-year update: nsf.gov/focus-areas/ai/nairr · [NAIRR at 2 years](#)
- NAIRR-OC solicitation (transition to permanent program): [NSF NAIRR-OC](#)
- Provider documentation: [Purdue Anvil NAIRR](#) · [NCSA Delta allocations](#) · [NCSA Delta usage report](#)
- Industry contributions: [NVIDIA \\$30M / Azure \\$20M](#) · [Microsoft Research NAIRR](#)
- Hardware specs cross-checked against [NVIDIA H100/H200/GH200 datasheets](#), [Intel Data Center GPU Max 1550 brief](#), and [Azure ND H100/H200 v5 product pages](#).